

Abdullah Mahmoud Abdullah

Data Engineer

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PROFESSIONAL SUMMARY

Junior Data Engineer focusing on scalable ETL pipelines, cloud integration, and AI-driven data systems. Strong technical skills in managing large datasets, with great teamwork; frequently leads weekly meetings to share progress, resolve issues, and ensure the team meets its main business goals.

SKILLS

- Programming & Scripting:** Python (Pandas), SQL, Bash/Shell Scripting
- Cloud Platforms:** AWS Solutions Architecture (S3, Lambda, Glue, Athena, EC2, EventBridge, SNS), Snowflake
- Data Engineering & Big Data:** ETL/ELT Pipelines, Apache Spark, Hadoop, Apache Airflow, Apache Kafka
- Databases:** Relational Databases (PostgreSQL, RDBMS), NoSQL, Data Warehousing
- Business Intelligence & Visualization:** Looker Studio, Streamlit, IBM Cognos Analytics
- Machine Learning & AI:** Generative AI (Google Gemini, Groq API), Predictive Modelling, NLP (Whisper V3)
- Soft Skills:** Structured Problem Solving, Pyramid Principle Communication, Adaptability, Agile Collaboration

EXPERIENCE

- Roto House | AI Engineer** Oct 2025 – Present
- Developed a factory management web app with an AI-driven backend (Node.js, PostgreSQL) to automate reporting and digitize industrial workflows, streamlining operations for 68 active technicians.
 - Processed 20+ daily audio using FFmpeg and Whisper V3 (via Groq API) for accurate speech-to-text transcription.
 - Leveraged Google Gemini to analyze visual workflows, extract key operational data, and auto-populate target forms.

PROJECTS

- End-to-End Data Engineering Platform (IBM Capstone)**
- Architected relational and NoSQL databases to handle diverse data types securely and efficiently.
 - Built and orchestrated scalable ETL pipelines utilizing Python, Shell Scripting, and Apache Airflow.
 - Leveraged Apache Spark to process large datasets, integrating the final data warehouse with Looker Studio for interactive dashboards.

- Global Weather Analytics Pipeline**
- Engineered an automated ETL pipeline using Python and Apache Airflow to schedule and extract daily weather data from the Open-Meteo API.
 - Transformed the raw, nested JSON payloads into structured, analysis-ready datasets utilizing Pandas.
 - Developed an interactive Streamlit dashboard to visualize weather trends.

- Orchestrating an IMDb ELT Pipeline: AWS to Snowflake Integration**
- Built an automated AWS ELT pipeline using EventBridge, Lambda, and SNS to extract over 670 MB IMDb datasets.
 - Optimized Snowflake ingestion using CDC and SQL MERGE, eliminating full-loads to achieve data availability.
 - Developed a Streamlit dashboard powered by Snowflake, enabling live data visualization via SQL pushdown.

- YouTube AWS Serverless Data Pipeline**
- Engineered a data pipeline, utilizing AWS Lambda for the extraction of raw JSON and CSV files from Amazon S3.
 - Processed and optimized datasets using AWS Glue by partitioning data and converting files to Parquet format.
 - Centralized the transformed data within Snowflake and developed dashboards using Looker Studio and Streamlit.

CERTIFICATIONS

- IBM Data Engineering Professional Certificate** | Coursera | Mar 2026
- Snowflake Data Engineering Professional Certificate** | Coursera | Apr 2026
- AWS Cloud Solutions Architect** | Coursera | May 2026

EDUCATION

- Bachelor of Computer Science** | Faculty of Computers and Artificial Intelligence | Sep 2022 - Jul 2026
- Relevant Coursework:** Data Modeling, Data Architecture, Database Management, Artificial Intelligence.